Python Tip: the `in` Operator

Need to check if something is contained in a list?

Want to see if a string contains a substring?

Use the `in` operator for a clean, readable one-liner!

Checking Lists

Use `in` to check if a value exists in a list.

```
fruits = ["apple", "banana", "cherry"]

# Don't do this
found = False
for fruit in fruits:
    if fruit == "banana":
        found = True
print(found)

# Do this instead
print("banana" in fruits)
print("grape" in fruits)
```

```
True
True
False
```

Checking Strings

`in` also works on strings to check for substrings.

```
message = "Hello, world!"

print("world" in message)
print("python" in message)
```

```
True
False
```

Checking Dictionaries

With dictionaries, `in` checks the **keys** by default.

```
scores = {"alice": 95, "bob": 87, "carol": 92}
# Check keys
print("alice" in scores)
# Check values
print(95 in scores.values())
# Check pairs
print(("bob", 87) in scores.items())
```

```
True
True
True
```

Using `not in`

Flip the check with `not in` to test for absence.

```
banned = ["spam", "ads", "clickbait"]

username = "spam"
if username not in banned:
    print(f"Welcome, {username}!")
else:
    print(f"{username} is banned.")
```

```
spam is banned.
```

Wrap-Up

Now you can use `in` and `not in` to check membership in lists strings, and dictionaries cleanly and concisely!

Follow me for more tips.

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